Choudhary Manish

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EDUCATION

Sardar Vallabhbhai National Institute of Technology, Surat, Gujarat **CGPA: 9.03/10**
Integrated M.Sc. in Mathematics *11/2022 – Expected Graduation: 2027*

Relevant Coursework: Data Structures & Algorithms | Database Management Systems | Operating Systems | Computer Networks | Artificial Intelligence | Data Science | Fuzzy Set Theory | Probability & Statistics | Linear Algebra

EXPERIENCE

Research Intern – SAC ISRO *05/2026 – 07/2026*
Bundle Block Adjustment for Satellite Image Orientation *Ahmedabad, Gujarat*

- Developed core C++ modules for a Bundle Block Adjustment pipeline for stereo orbital imagery, contributing to the design and implementation of a satellite image orientation workflow.
- Designed and implemented coordinate transformation modules across orbit, attitude, and camera look-angle reference frames, using rotation matrix models to establish satellite image geometry.
- Developed and integrated modules for tie-point processing, space intersection, and non-linear least-squares optimization using the Ceres Solver library.
- Processed real satellite orbit, attitude, and tie-point datasets to construct, validate, and test the geometric sensor model end-to-end.

Research Intern – Indian Institute of Technology (IIT), Jodhpur *05/2025 – 07/2025*
Stock Trend Analytics *Jodhpur, Rajasthan*

- Developed a data-driven stock trend analytics system using Python, technical indicators, and fuzzy inference methods on historical market datasets.
- Analyzed volatility and trend indicators to identify patterns and generate Buy/Sell/Hold signals for data-driven decision-making.
- Evaluated model performance using Python and translated analytical results into actionable insights for interpreting market trends.

PROJECTS

Packet Analyzer – Deep Packet Inspection Engine

- Developed a packet analysis engine to parse network traffic and inspect protocols across multiple network layers.
- Implemented deep packet inspection techniques to extract TLS SNI metadata and classify network applications.
- Designed network-flow tracking and rule-based filtering to identify and selectively block traffic based on configurable criteria.

Sentiment Analysis (NLP Project)

- Developed a rule-based NLP sentiment analysis system using Triangular Fuzzy Numbers (TFNs) to model linguistic ambiguity, with support for negation, modifiers, and compound sentences.
- Designed and implemented a fuzzy inference engine with custom sentiment lexicons, text preprocessing, and defuzzification for nuanced sentiment classification.
- Tested the system on 105 curated cases and achieved 74.29% classification accuracy.

SKILLS

Programming: C++ | C | SQL | Python | MATLAB
Data Analytics & ML: Data Analysis | Data Preprocessing | NumPy | Pandas | scikit-learn

ACHIEVEMENTS

- Qualified GATE 2026 (Data Science & AI).
- Organized "Memefy Math" during National Mathematics Week, coordinating an event that attracted 100+ participants.

POSITIONS OF RESPONSIBILITY

Head of Creative & Design Committee – Chintan Pratishthan, SVNIT *2023 – 2025*
Designed 20+ promotional assets for institute events and coordinated the annual Prashikshan Varg camp with 50+ student participants.

Executive – Association for Computing Machinery, SVNIT *2023 – 2024*
Supported publicity and coordination for technical workshops and hackathons, collaborating with student teams.

CERTIFICATIONS

SQL (Intermediate) | **Scientific Computing with Python** | **Data Structures & Algorithms (C++)** | **Power BI for Beginners**